

Lab07 – Labels and annotations

In this exercise, you will exercise assigning labels and annotations to a set of Pods. Moreover, you will use `kubectl` to query for Pods based on different requirements.

1. Create three different Pods with the names `frontend`, `backend` and `database` that use the image `nginx`. Declare labels for those Pods as follows:
 - `frontend`: `env=prod`, `team=shiny`
 - `backend`: `env=prod`, `team=legacy`, `app=v1.2.4`
 - `database`: `env=prod`, `team=storage`

```
vagrant@k8s-control-plane:~/lab$ kubectl run frontend --image=nginx --restart=Never --labels=env=prod,team=shiny -oyaml > frontend.yaml
vagrant@k8s-control-plane:~/lab$ kubectl run backend --image=nginx --restart=Never --labels=env=prod,team=legacy,app=v1.2.4 -oyaml > backend.yaml
vagrant@k8s-control-plane:~/lab$ kubectl run database --image=nginx --restart=Never --labels=env=prod,team=storage -oyaml > database.yaml
vagrant@k8s-control-plane:~/lab$ ls
backend.yaml database.yaml frontend.yaml
vagrant@k8s-control-plane:~/lab$ kubectl get pods --show-labels
NAME          READY   STATUS    RESTARTS   AGE   LABELS
backend       1/1     Running   0           80s   app=v1.2.4,env=prod,team=legacy
database      1/1     Running   0           49s   env=prod,team=storage
frontend      1/1     Running   0           2m9s   env=prod,team=shiny
vagrant@k8s-control-plane:~/lab$
```

2. Edit the existing Pods with the `edit` command and add the annotations as follows:
 - `frontend`: `contact=John Doe`, `commit=2d3mg3`
 - `backend`: `contact=Mary Harris`

```
vagrant@k8s-control-plane:~/lab$ vim frontend.yaml
vagrant@k8s-control-plane:~/lab$ cat frontend.yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    env: prod
    team: shiny
  annotations:
    commit: 2d3mg3
    contact: John Doe
  name: frontend
spec:
  containers:
  - image: nginx
    name: frontend
vagrant@k8s-control-plane:~/lab$ vim backend.yaml
vagrant@k8s-control-plane:~/lab$ cat backend.yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: v1.2.4
    env: prod
    team: legacy
  annotations:
    contact: 'Mary Harris'
  name: backend
spec:
  containers:
  - image: nginx
    name: backend
vagrant@k8s-control-plane:~/lab$
```

```
vagrant@kubernetes-control-plane:~/lab$ kubectl apply -f frontend.yaml
Warning: resource pods/frontend is missing the kubect.kubernetes.io/last-applied-configuration annotation which is required by kubectl apply.
kubectl apply should only be used on resources created declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
pod/frontend configured
vagrant@kubernetes-control-plane:~/lab$ kubectl apply -f backend.yaml
Warning: resource pods/backend is missing the kubect.kubernetes.io/last-applied-configuration annotation which is required by kubectl apply.
kubectl apply should only be used on resources created declaratively by either kubectl create --save-config or kubectl apply. The missing annotation will be patched automatically.
pod/backend configured
vagrant@kubernetes-control-plane:~/lab$
```

3. Use label selectors on the command line to query for all production Pods that belong to the teams `shiny` and `legacy`.

```
vagrant@kubernetes-control-plane:~/lab$ kubectl get pods -l 'team in (shiny, legacy)',env=prod --show-labels
NAME      READY   STATUS    RESTARTS   AGE   LABELS
backend    1/1     Running   0           9m7s   app=v1.2.4,env=prod,team=legacy
frontend   1/1     Running   0           9m56s   env=prod,team=shiny
vagrant@kubernetes-control-plane:~/lab$
```

4. Remove the label `env` from the `backend` Pod and rerun the selection.

```
vagrant@kubernetes-control-plane:~/lab$ kubectl label pod backend env-
pod/backend unlabeled
vagrant@kubernetes-control-plane:~/lab$ kubectl get pods -l 'team in (shiny, legacy)',env=prod --show-labels
NAME      READY   STATUS    RESTARTS   AGE   LABELS
frontend   1/1     Running   0           14m   env=prod,team=shiny
vagrant@kubernetes-control-plane:~/lab$
```

5. Render the surrounding 3 lines of YAML of all Pods that have annotations

```
vagrant@kubernetes-control-plane:~/lab$ kubectl get pods -o yaml | grep -C 3 'annotations:'
- apiVersion: v1
  kind: Pod
  metadata:
    annotations:
      contact: Mary Harris
      kubect.kubernetes.io/last-applied-configuration: |
        {"apiVersion":"v1","kind":"Pod","metadata":{"annotations":{"contact":"Mary Harris"},"labels":{"app":"v1.2.4","env":"prod","team":"legacy"},"name":"backend","namespace":"default"},"spec":{"containers":[{"image":"nginx","name":"backend"}]}}
  --
- apiVersion: v1
  kind: Pod
  metadata:
    annotations:
      commit: 2d3mg3
      contact: John Doe
      kubect.kubernetes.io/last-applied-configuration: |
vagrant@kubernetes-control-plane:~/lab$
vagrant@kubernetes-control-plane:~/lab$
```